Translating Cancer Research to the Clinic

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1. Motivation

- Oral Cancer is on the **rise** in the UK, with **12000** new cases and **4000** deaths per year. Low survival rate is attributed to **delayed** diagnosis [1].
- Currently, histopathologists identify malignancy in oral lesions using optical images of biopsies. Tracking patient outcomes reveals that the accuracy of their predictions is **$\sim$40%.** [2]
- Using the FTIR (Fourier-Transform infrared) spectra of the biopsies and machine learning (ML), an identification method has been developed to identify malignant (M) and benign (B) tumours with **$>$90%** accuracy [2].
- The FTIR data is taken using CaF_2 slides which are harder to handle than the glass slides histopathologists are familiar with.

Can FTIR data taken on glass be used, despite the loss of part of the spectrum due to the absorbance of glass?

2. Biopsy Imaging Technique

- The range of the FTIR spectrum that is lost is the fingerprint region.
- This contains a wealth of information on the bio-molecules present in the cells.
- Can the functional region, visible on glass, provide the information necessary to identify malignancy?

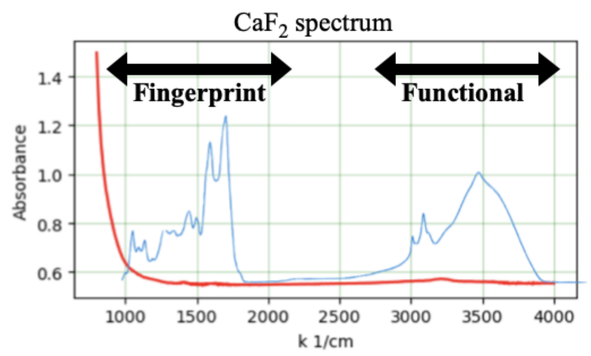

Figure 1: CaF_2 absorbance (red), with IR spectrum from pixel in datacube for one biopsy (blue) taken on CaF_2.

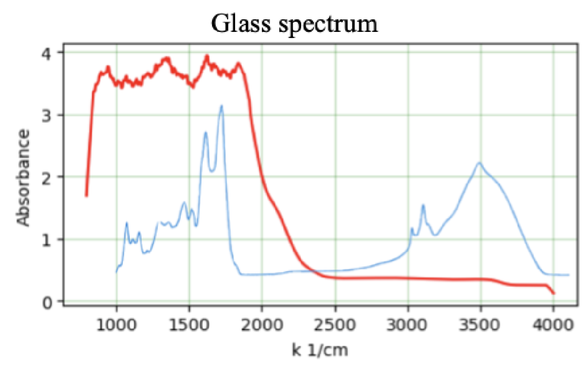

Figure 2: Glass absorbance (red), illustrating loss of fingerprint region if spectra taken on glass.

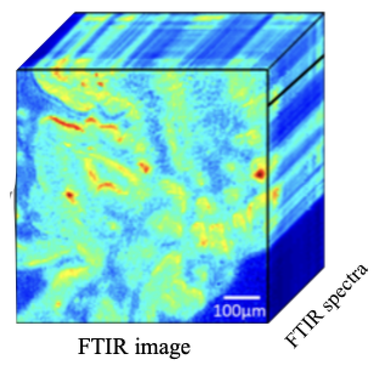

Figure 3: FTIR datacube: each image pixel contains a complete FTIR spectrum.

3. ML Analysis

Original Analysis:

- A pathologist identifies lesions.
- Take FTIR images of biopsy, to obtain datacube across all spectra (Figure 3).
- Using absorbance values in the fingerprint region (Figure 1), **train** a multi layer perceptron (MLP) to identify normal, M or B cells.
- **Test** using independent set of pixels.

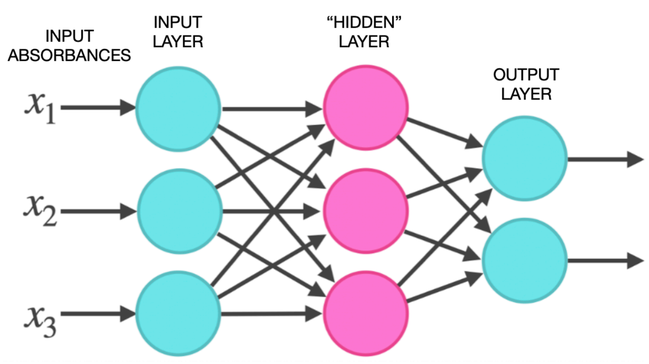

Figure 4: Structure of a MLP.

Evaluate true positive and true negative rate:

$$TPR = \frac{TP}{TP+FN} \qquad TNR = \frac{TN}{TN+FP}$$

Clinic-oriented analysis:

- Repeat the above original analysis, but using the **functional** region of the spectrum (Figure 2), not absorbed by glass.

4. Results

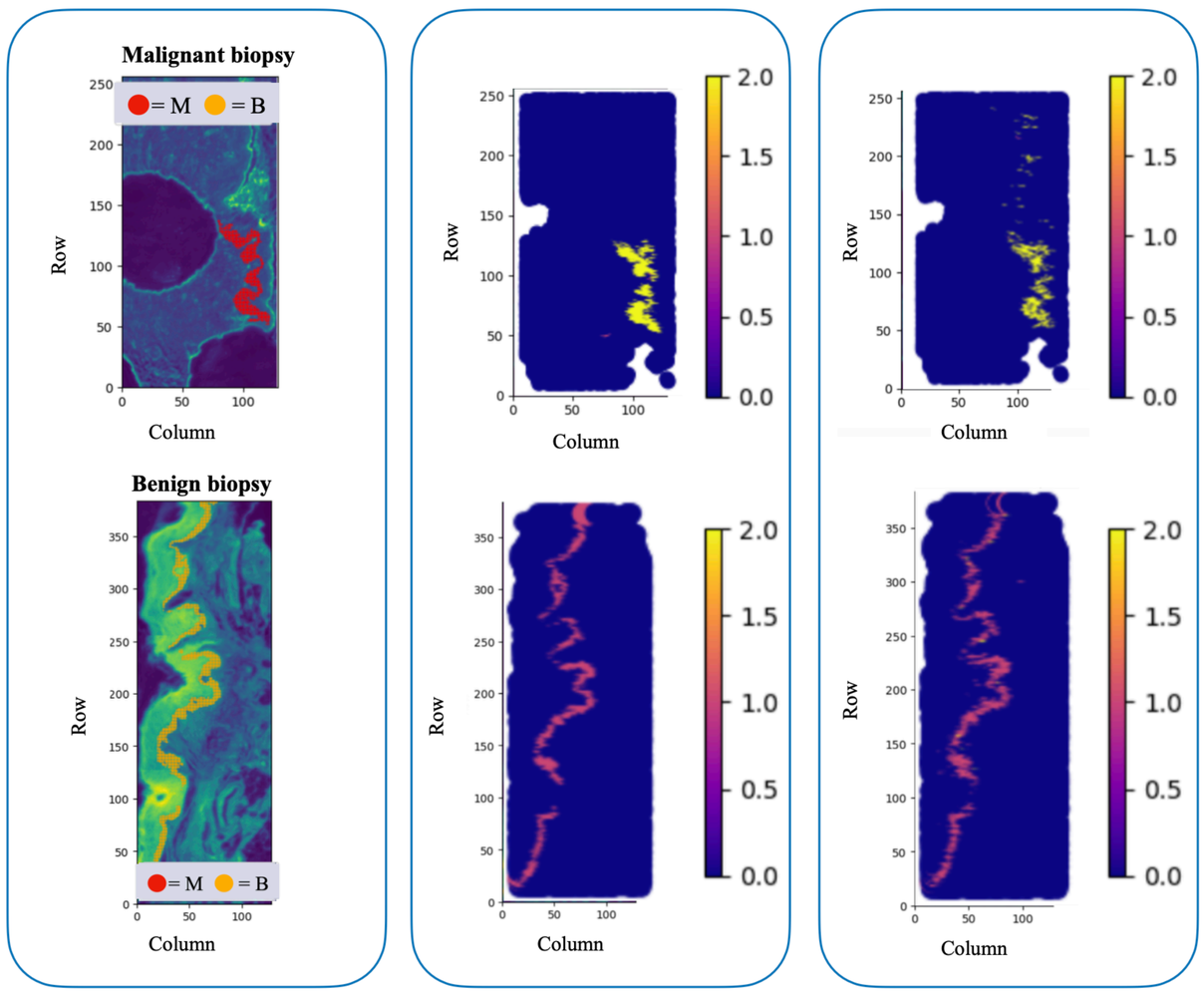

Figure 5: Cells identified as M or B (left). Original analysis is run and test results of the MLP indicate whether M or B areas have been found (middle). The clinic-oriented analysis is run and test results shown (right). Colourbar shows cell type: 0 = normal tissue, 1 = B, 2 = M.

Fingerprint region

	TPR	TNR
M	$\sim$0.93	$\sim$0.99
B	$\sim$0.96	$\sim$0.99

Functional region

	TPR	TNR
M	$\sim$0.79	$\sim$0.97
B	$\sim$0.81	$\sim$0.98

5. Conclusion and further work

1. Glass slides **can be used** to accurately determine the prognosis of a tumour.
2. Results are **significantly better** than is currently seen in the clinic.
3. Optimisation of the ML algorithms using the data on glass is ongoing.

References:

[1] Ramachandran, S (2024). *Oral Oncology Reports* 14, 16050

[2] Al Jedani, S.*et al* (2024). *Sci Rep* 12. 100678